EECS 4422 Computer Vision
Frequency-Domain Deblurring

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Implementing and comparing Fourier-domain restoration methods for deblurring

Image source

Clean reference images are taken from **CBSD68** (68 colour natural images, 321 × 481), obtained from the benchmark collection at https://github.com/majedelhelou/denoising_datasets.

Degradation model | Synthetically blurring the input images

Following the course notation, g denotes the original image and f the degraded observation:

$$f(x,y) = h(x,y) * g(x,y) + n(x,y) \tag{1}$$
$$F(u,v) = H(u,v)\,G(u,v) + N(u,v) \tag{2}$$

g = original (sharp) image, h = blur kernel (PSF), n = additive noise, f = observed blurry image; capitals denote the corresponding 2-D discrete Fourier transforms, and $\hat{G}$ denotes the restored estimate of G.

What the blurring operation does

Applies a convolution:

$$(h * g)(x,y) = \sum_i \sum_j h(i,j)\, g(x-i,\ y-j) \tag{3}$$

The kernel used here is an isotropic Gaussian of width σ, normalised to unit sum:

$$h(x,y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2+y^2}{2\sigma^2}\right), \qquad \sum_{x,y} h(x,y) = 1 \tag{4}$$

Because the weights sum to one, the local average is preserved, so blurring redistributes intensity without changing overall brightness. The kernel is evaluated on an odd window of half-width $\lceil 3\sigma \rceil$, beyond which its values are negligible.
The filter is applied separately.

Inverse filter

By the convolution theorem, convolution in the spatial domain becomes multiplication in the frequency domain:

$$g * h \iff G\,H \tag{5}$$

Applying this to the degradation model turns the convolution into a product:

$$f = h * g + n \iff F = H\,G + N \tag{6}$$

Considering the Fourier transforms of g, f and h, (we assume noise $= 0$) :

$$G(u,v) = \text{the sharp image to recover,} \tag{7}$$
$$F(u,v) = \text{the blurred image} \tag{8}$$
$$H(u,v) = \text{the blur, as a gain applied to each frequency.} \tag{9}$$

Blurring multiplies G by H, so the restoration divides it back out:

$$\hat{G} = \frac{F}{H} \tag{10}$$

Since $|H|$ becomes vanishingly small at high frequencies, it is floored at ε before dividing (to avoid division by 0):

$$\hat{G}(u,v) = \frac{F(u,v)}{H_\varepsilon(u,v)}, \qquad H_\varepsilon = \begin{cases} H, & |H| \geq \varepsilon, \\ \varepsilon, & |H| < \varepsilon, \end{cases} \qquad \varepsilon = 10^{-2} \tag{11}$$

Steps

1. Build the kernel h from σ, then transform it: $H = \mathcal{F}\{h\}$. A Gaussian is fully determined by σ, so no other information is needed, and this is the identical H that produced the blur.
2. Floor it: $H_\varepsilon = \max(|H|,\ \varepsilon)$, keeping H elsewhere.
3. For each colour channel:
   (a) $F = \mathcal{F}\{f\}$ (pixels → frequencies)
   (b) $\hat{G} = F/H_\varepsilon$ (divide out the blur)
   (c) $\hat{g} = \mathrm{Re}\ \mathcal{F}^{-1}\{\hat{G}\}$ (frequencies → pixels)
   (d) keep the real part of output, discard the imaginary part,

Wiener filter

Rather than assume $n = 0$, find the linear shift-invariant filter w minimising the expected mean square error between the estimate $\hat{g} = w * f$ and the original image g:

$$\min_w \mathbb{E}\left[\iint \left(\hat{g}(x,y) - g(x,y)\right)^2 dx\, dy\right] \tag{12}$$

The solution is expressed in terms of *power spectral densities* (PSDs) — the expected energy each signal carries at each frequency — of the original image, the noise, and the degraded image respectively:

$$P_g(u,v) = \mathbb{E}\left[|G(u,v)|^2\right] \tag{13}$$

$$P_n(u,v) = \mathbb{E}\left[|N(u,v)|^2\right] \tag{14}$$

$$P_f(u,v) = \mathbb{E}\left[|F(u,v)|^2\right] \tag{15}$$

Substituting $F = HG + N$ into $P_f = \mathbb{E}\left[|F|^2\right]$ and expanding with $|a+b|^2 = |a|^2 + |b|^2 + 2\,\mathrm{Re}\,(ab^*)$:

$$\begin{aligned} P_f &= \mathbb{E}\left[|HG + N|^2\right] \\ &= |H|^2\,\mathbb{E}\left[|G|^2\right] + \mathbb{E}\left[|N|^2\right] + 2\,\mathrm{Re}\,\mathbb{E}[HGN^*] \end{aligned} \tag{16}$$

The cross term vanishes because H is deterministic, G and N are independent, and the noise has zero mean:

$$\mathbb{E}[HGN^*] = H\,\mathbb{E}[GN^*] = H\,\mathbb{E}[G]\,\mathbb{E}[N^*] = 0 \tag{17}$$

Thus:

$$P_f = |H|^2\,P_g + P_n \tag{18}$$

The best W is the one that leaves an error independent of the data: if the error still depended on the data, adjusting W could remove more of it.

$$\mathbb{E}[(WF - G)\,F^*] = 0 \tag{19}$$

Rearranging, then substituting $F = HG + N$ and dropping the cross term as before:

$$W = \frac{\mathbb{E}[GF^*]}{\mathbb{E}\left[|F|^2\right]} = \frac{H^* P_g}{P_f} = \frac{H^* P_g}{|H|^2\,P_g + P_n} \tag{20}$$

That is, $W \approx 1/H$ where the signal dominates and $W \approx 0$ where the noise does. Writing $K = P_n/P_g$:

$$\hat{G}(u,v) = \frac{H^*(u,v)}{|H(u,v)|^2 + K}\,F(u,v), \qquad K = 3 \times 10^{-4} \tag{21}$$

The denominator cannot reach zero, so no ε-floor is needed; $K \to 0$ recovers the inverse filter.

Steps

1. Build the kernel h from σ, then transform it: $H = \mathcal{F}\{h\}$.
2. Form the filter once: $W = H^*/\left(|H|^2 + K\right)$.
3. For each colour channel:
   (a) $F = \mathcal{F}\{f\}$ (pixels → frequencies)
   (b) $\hat{G} = W\,F$ (apply the filter)
   (c) $\hat{g} = \mathrm{Re}\,\mathcal{F}^{-1}\{\hat{G}\}$ (frequencies → pixels)

Note: both the estimation of $K = P_n/P_g$ and the actual oracle values from original image are used in comparison in the functions `wiener_filter` and `wiener_filter_oracle`.

Constrained least squares (Laplacian)

The Wiener filter needs P_n and P_g, which are unknown in general. Constrained least squares replaces those statistics with a preference: among all estimates consistent with the data, choose the smoothest. Roughness is measured by the Laplacian, ∇^2, the sum of the second derivative along each axis:

$$\nabla^2 g = \frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2} \tag{22}$$

Differencing twice along x gives the discrete form of each term:

$$\begin{aligned}\frac{\partial^2 g}{\partial x^2} &\approx \big[g(x{+}1) - g(x)\big] - \big[g(x) - g(x{-}1)\big] \\ &= g(x{+}1) - 2g(x) + g(x{-}1)\end{aligned} \tag{23}$$

so each axis contributes the kernel $[\,1\ \ {-2}\ \ 1\,]$, and their sum is the 3×3 operator p:

$$p = \begin{bmatrix}0 & 0 & 0\\ 1 & -2 & 1\\ 0 & 0 & 0\end{bmatrix} + \begin{bmatrix}0 & 1 & 0\\ 0 & -2 & 0\\ 0 & 1 & 0\end{bmatrix} = \begin{bmatrix}0 & 1 & 0\\ 1 & -4 & 1\\ 0 & 1 & 0\end{bmatrix} \tag{24}$$

Its weights sum to zero, so a region of constant intensity gives exactly zero response: the operator measures curvature only, and is blind to brightness.

Minimise that roughness subject to the estimate reproducing the observed image:

$$\min_{\hat{g}} \ \|p * \hat{g}\|^2 \qquad \text{subject to} \qquad \|f - h * \hat{g}\|^2 = \|n\|^2 \tag{25}$$

Introducing a multiplier γ for the constraint and setting the derivative of the combined objective to zero gives, in the frequency domain:

$$\hat{G}(u,v) = \frac{H^*(u,v)}{|H(u,v)|^2 + \gamma\,|P(u,v)|^2}\, F(u,v), \qquad \gamma = 10^{-4} \tag{26}$$

where $P = \mathcal{F}\{p\}$. This is the Wiener filter with the constant K replaced by $\gamma\,|P|^2$. Because p is high-pass, that penalty varies with frequency:

$$P(0,0) = 0 \quad \text{the mean is unconstrained,} \tag{27}$$

$$|P|^2 \to 64 \quad \text{maximal penalty at the highest frequencies.} \tag{28}$$

So the damping is applied exactly where the inverse filter fails, and no noise statistics are required — only the assumption that the original image is smooth.

Steps

1. Build the kernel h from σ and transform it: $H = \mathcal{F}\{h\}$.
2. Build the Laplacian p on the same grid and transform it: $P = \mathcal{F}\{p\}$.
3. Form the filter once: $W = H^* / \left(|H|^2 + \gamma\,|P|^2\right)$.
4. For each colour channel:
   (a) $F = \mathcal{F}\{f\}$ (pixels $\to$ frequencies)
   (b) $\hat{G} = W\,F$ (apply the filter)
   (c) $\hat{g} = \mathrm{Re}\ \mathcal{F}^{-1}\{\hat{G}\}$ (frequencies $\to$ pixels)

Results

All four filters are run over the same 68 CBSD68 images, blurred with $\sigma = 2$ and written to 8-bit PNG, so the only noise present is the quantisation of that round trip. Each restored image is clipped to $[0, 1]$ before scoring. PSNR measures squared error; SSIM compares local mean, contrast and correlation over a 7×7 window, so the two disagree where a method trades error for structure.

Method	PSNR (dB)	Δ PSNR	SSIM	ms/image
Blurry input	24.29	—	0.6703	—
Inverse filter	19.50	-4.79	0.3454	24
Wiener filter	27.84	+3.56	0.8242	24
Wiener filter (oracle)	28.19	+3.90	0.8259	36
Constrained least squares	27.84	+3.55	0.8263	27

Table 1. Mean PSNR and SSIM over 68 images, $\sigma = 2$.

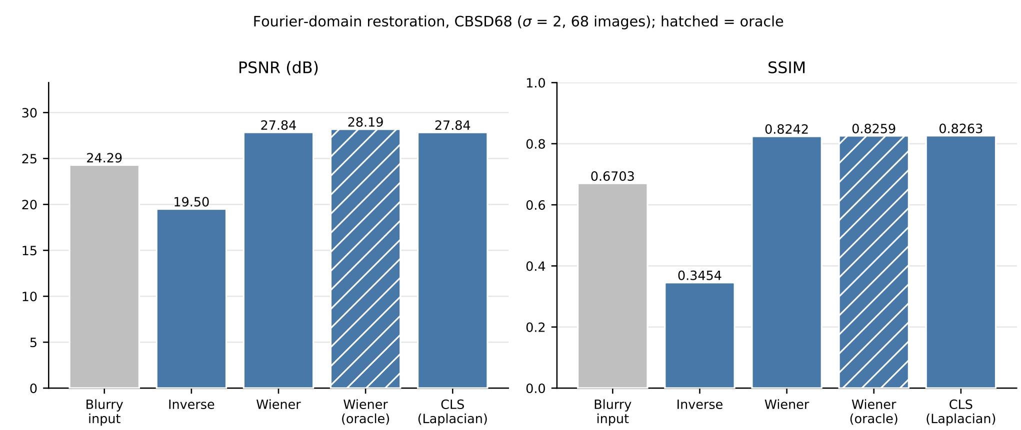


Figure 1. The same figures. The oracle (hatched) uses the ground-truth P_g, so it is an upper bound rather than an achievable result.

Sample restorations

One grid per image, each cell captioned with its own PSNR and SSIM against the ground truth. Generated by `save_panels()`, so the scores below cannot drift from those in Table 1.

Blurry input — 32.95 dB / 0.9526

Inverse filter — 19.99 dB / 0.1362

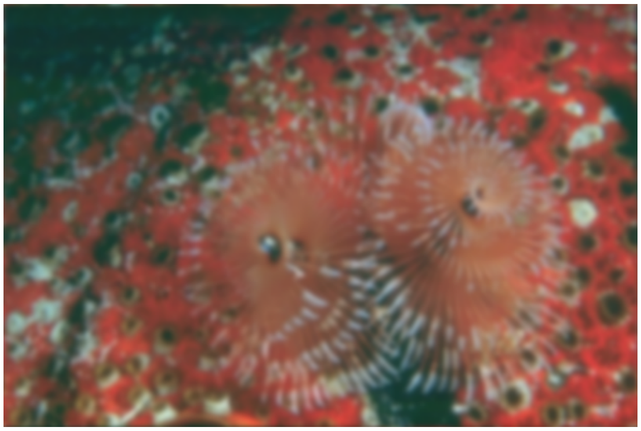
Wiener filter — 36.01 dB / 0.8970

Wiener filter (oracle) — 37.12 dB / 0.9514

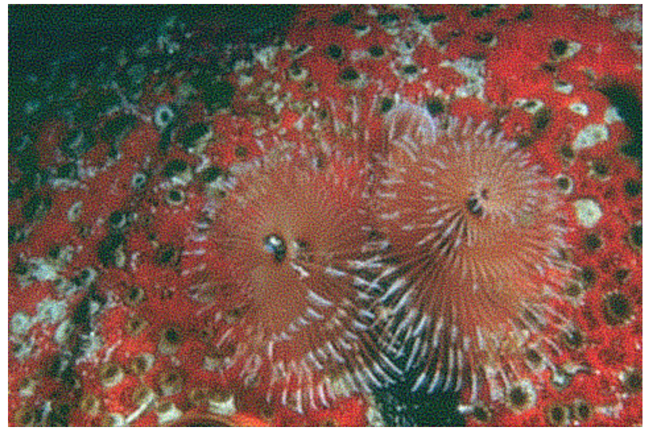
Constrained least squares — 36.22 dB / 0.9073

Ground truth

Figure 2. `3096.png` restored from $\sigma = 2$ blur. Read left to right, top to bottom.

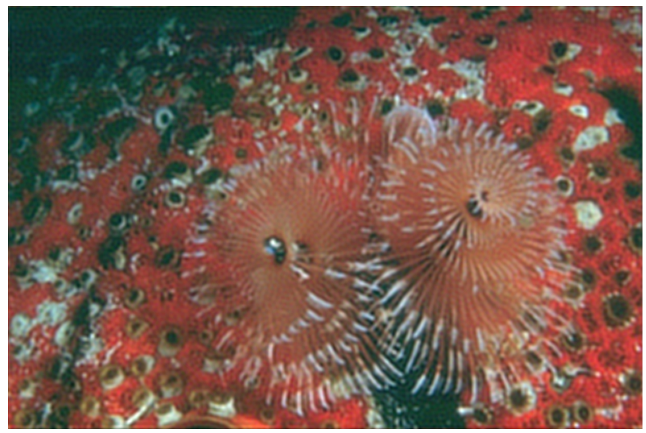


Blurry input — 24.35 dB / 0.6417



Inverse filter — 19.46 dB / 0.4008

Wiener filter — 29.22 dB / 0.8687



Wiener filter (oracle) — 29.48 dB / 0.8699

Constrained least squares — 29.19 dB / 0.8693

Ground truth

Figure 3. `12084.png` restored from $\sigma = 2$ blur. Read left to right, top to bottom.

Blurry input — 24.71 dB / 0.6603

Inverse filter — 19.47 dB / 0.3559

Wiener filter — 28.51 dB / 0.8353

Wiener filter (oracle) — 28.72 dB / 0.8373

Constrained least squares — 28.50 dB / 0.8367

Ground truth

Figure 4. `16077.png` restored from $\sigma = 2$ blur. Read left to right, top to bottom.

Blurry input — 23.94 dB / 0.6570

Inverse filter — 19.45 dB / 0.3494

Wiener filter — 27.68 dB / 0.8250

Wiener filter (oracle) — 27.88 dB / 0.8203

Constrained least squares — 27.66 dB / 0.8266

Ground truth

Figure 5. `19021.png` restored from $\sigma = 2$ blur. Read left to right, top to bottom.

Blurry input — 21.68 dB / 0.7534

Inverse filter — 19.82 dB / 0.4414

Wiener filter — 27.76 dB / 0.8996

Wiener filter (oracle) — 28.21 dB / 0.8835

Constrained least squares — 27.75 dB / 0.9015

Ground truth

Figure 6. `24077.png` restored from $\sigma = 2$ blur. Read left to right, top to bottom.